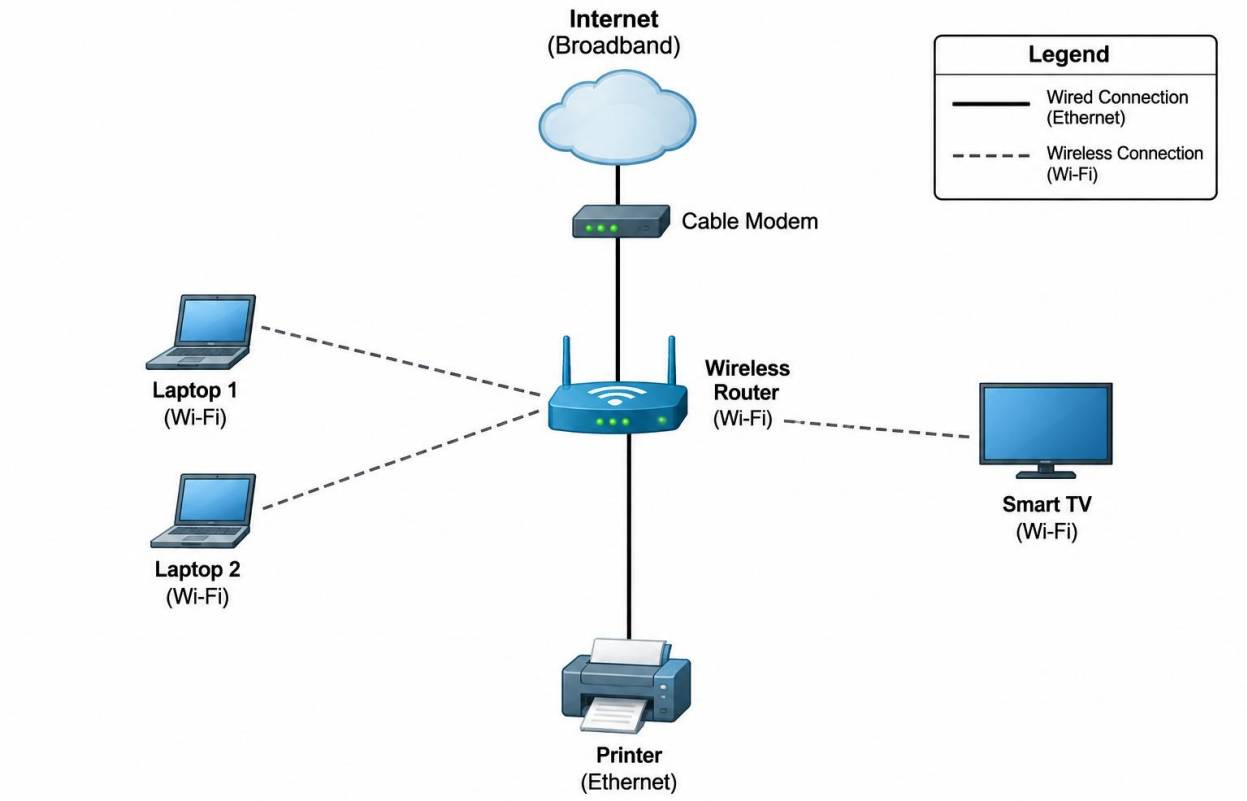
1.Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Ans.



2) List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Ans. **Hardware Components Required for SOHO Network**

No.	Hardware component	Suggested brand/model
1.	Wi-Fi Router	TP-Link Archer C6
2.	Broadband Modem/ONT	Airtel or Jio ONT
3.	Laptop	Any 2 laptops (HP/Dell)
4.	Smart TV	Samsung / LG Smart TV

5. Printer HP Wi-Fi Printer
6. LAN Cable D-Link Cat6 Ethernet Cable
7. Network Switch TP-Link TL-SG1005D (optional)
8. Wi-Fi Extender TP-Link RE200 (optional)

Other Accessories

- RJ45 connectors
- Router power adapter
- Ethernet cables for wired connections

3) Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Ans. Steps to Configure a Wi-Fi Router for SOHO Network

1. Connect the Router: Connect the router to the broadband modem using an Ethernet cable.

2. Open Router Settings: Connect your laptop to the router and open a browser. Enter 192.168.0.1 or 192.168.1.1.

3. Login: Enter the default username and password given on the router label.

4. Set Wi-Fi Name (SSID): Go to Wireless Settings and create a unique Wi-Fi name, such as My_SOHO_Network.

5. Enable WPA2 Security: Select WPA2-PSK (AES) security and set a strong Wi-Fi password.

6. Change Admin Password: Open System Settings and change the default admin password to a new strong password.

7. Save Settings: Click Save or Apply and restart the router if required.

8. Connect Devices: Connect both laptops, smart TV and printer to the new Wi-Fi network.

4) Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Ans.

Device	MAC Address (Example)	IP Address	Type
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	**Static**
Laptop 1	00:1A:2B:3C:4D:02	192.168.1.100	**DHCP**
Laptop 2	00:1A:2B:3C:4D:03	192.168.1.101	**DHCP**
Smart TV	00:1A:2B:3C:4D:04	192.168.1.10	**Static**
Printer	00:1A:2B:3C:4D:05	192.168.1.20	**Static**

Steps:

1. Open the router settings.
2. Enable DHCP for laptops.
3. Assign a fixed IP address to the printer.
4. Assign a fixed IP address to the smart TV.
5. Save the settings and restart the devices.

5) Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Ans. Troubleshooting Internet Connection Problems in SOHO Network

If one laptop cannot connect to the internet, I will check the following problems:

Possible Cause	Diagnosis and Solution
1. Wi-Fi is turned OFF	Check Wi-Fi settings and turn it ON.
2. Wrong Wi-Fi Password	Enter the correct Wi-Fi password and reconnect.
3. IP Address Problem	Restart the laptop or set IP settings to automatic (DHCP).
4. Router Problem	Restart the Wi-Fi router and check the internet connection.
5. Network Driver Problem	Update or reinstall the Wi-Fi driver.

Steps to Troubleshoot:

- **Check whether Wi-Fi is enabled.**
- **Restart the laptop.**
- **Check the Wi-Fi password.**
- **Restart the router.**
- **Test the internet on another device.**